Parabolas: Focus & Directrix

Conic Sections — one parabola, four directions

The big idea

A **parabola** is every point that is the **same distance** from a point (the **focus**) and a line (the **directrix**). The **vertex** is halfway between them, and p is the distance (with a sign) from the vertex to the focus.



opens up **opens down** **opens right** **opens left**

	Equation	Focus	Directrix	Axis
Up ($p > 0$) / Down ($p < 0$)	$(x-h)^2 = 4p(y-k)$	$(h,\ k+p)$	$y = k-p$	$x = h$
Right ($p > 0$) / Left ($p < 0$)	$(y-k)^2 = 4p(x-h)$	$(h+p,\ k)$	$x = h-p$	$y = k$

Simple case first. $x^2 = 4y$: here $4p = 4$, so $p = 1$. Vertex $(0,0)$, focus $(0,1)$, directrix $y = -1$. Check with the point $(2,1)$ on the curve: distance to the focus is 2, distance to the line $y = -1$ is 2. ✓

Watch out: The squared letter tells the direction. $x^2 \Rightarrow$ opens up or down; $y^2 \Rightarrow$ opens right or left. Then the sign of p decides which way. And $4p$ is not p: in $x^2 = 12y$, $p = 3$, not 12.

0 Warm-Up: Algebra 2 skills you'll need

MEDIUM

These come from earlier Algebra 2 units. Each one shows up again below.

1. **Distance formula.** Find the distance between $(-2, 5)$ and $(4, -3)$.

2. **Distance formula.** The point $(x, 4)$ is 5 units from $(1, 0)$. Find every possible x.

3. **Completing the square.** Write $y = x^2 + 6x + 5$ in vertex form and give the vertex.

4. **Completing the square.** Write $y = 2x^2 - 8x + 5$ in vertex form.

5. **Transformations.** Describe $g(x) = -\frac{1}{2}(x+1)^2 + 4$ compared with $y = x^2$, and give its vertex.

6. **Solving quadratics.** Solve $2x^2 - 3x - 5 = 0$.

7. **Linear–quadratic system.** Solve $y = x^2 - 2$ and $y = x + 4$.

8. **Inverse functions.** $f(x) = (x-1)^2 + 2$ for $x \geq 1$. Find $f^{-1}(x)$ and its domain. (Its graph is half of a sideways parabola.)

A Opens Up $(x-h)^2 = 4p(y-k),\ p>0$

Example · opens up

Find the vertex, focus and directrix of $(x-1)^2 = 4(y+2)$.
Vertex $(h,k) = (1,-2)$. $4p = 4$, so $p = 1$ (positive: opens up). Focus $(1,-2+1) = (1,-1)$.
Directrix $y = -2-1 = -3$.

9. EASY Find the vertex, focus and directrix of $x^2 = 8y$.

10. EASY Find the focus and directrix of $x^2 = 12y$.

11. EASY Write the equation of the parabola with vertex $(0,0)$ and focus $(0,5)$.

12. MEDIUM Find the vertex, focus, directrix and axis of symmetry of $(x-2)^2 = 8(y+1)$.

13. HARD Write $y = \frac{1}{4}x^2 - x + 3$ in the form $(x-h)^2 = 4p(y-k)$. Then find the vertex, focus and directrix.

B Opens Down $(x-h)^2 = 4p(y-k),\ p<0$

Example · opens down

Find the vertex, focus and directrix of $x^2 = -8y$.
$4p = -8$, so $p = -2$ (negative: opens down). Vertex $(0,0)$, focus $(0,-2)$, directrix $y = 2$ — the directrix is on the *opposite* side.

14. EASY Find the vertex, focus and directrix of $x^2 = -16y$.

15. EASY Find the focus and directrix of $x^2 = -4y$.

16. EASY Write the equation of the parabola with vertex $(0,0)$ and directrix $y = 3$.

17. MEDIUM Write the equation of the parabola with focus $(1,-2)$ and directrix $y = 4$.

18. HARD Write $x^2 + 6x + 8y + 1 = 0$ in standard form. Find the vertex, focus and directrix.

C Opens Right $(y-k)^2 = 4p(x-h),\ p>0$

Example · opens right

Find the vertex, focus and directrix of $y^2 = 12x$.
y is squared: it opens sideways. $4p = 12$, so $p = 3$ (positive: right). Vertex $(0,0)$, focus $(3,0)$, directrix $x = -3$ (a *vertical* line).

19. EASY Find the vertex, focus and directrix of $y^2 = 4x$.

20. EASY Find the focus and directrix of $y^2 = 20x$.

21. EASY Write the equation of the parabola with vertex $(0,0)$ and focus $(2,0)$.

22. MEDIUM Find the vertex, focus, directrix and axis of symmetry of $(y-3)^2 = 12(x+2)$.

23. HARD **Satellite dish.** A dish's cross-section is a parabola with vertex at the origin, opening right. The dish is 24 inches wide and 4 inches deep. How far from the vertex should the receiver (at the focus) be placed?

D Opens Left $(y-k)^2 = 4p(x-h),\ p<0$

Example · opens left

Find the vertex, focus and directrix of $(y+1)^2 = -4(x-2)$.
Vertex $(2,-1)$. $4p = -4$, so $p = -1$ (negative: left). Focus $(2-1,\ -1) = (1,-1)$. Directrix $x = 2+1 = 3$.

24. EASY Find the vertex, focus and directrix of $y^2 = -8x$.

25. EASY Find the focus and directrix of $y^2 = -24x$.

26. EASY Write the equation of the parabola with vertex $(0,0)$ and directrix $x = 7$.

27. MEDIUM Write the equation of the parabola with vertex $(3,-1)$ and focus $(1,-1)$. Give its directrix.

28. HARD Write $y^2 - 4y + 4x + 12 = 0$ in standard form. Find the vertex, focus and directrix, then check one point with the definition.